Supplementary report to "The Statistical Value of Adaptive Feature Queries": material from earlier stages of the study

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This report collects analyses from earlier stages of the study that the submitted paper no longer uses. They are retained because they document negative and null findings: restricted branching menus on independently fitted forests, controlled refinements on preference and personality data, calibration transfer across age groups, pilot-based calibration, label-cost crossovers, and a synthetic budget curve. Numbered references to theorems, equations and appendix sections point into the submitted paper. Nothing here is needed for the results of the paper. All results are reproducible from the released code and public data as described in the paper's reproducibility section.

S1 Earlier Experiments: an Incomplete Archive, Restricted Menus, Controlled Refinements and Costs

This appendix retains the experiments of the previous version of the study. Their tasks, splits and summaries were specified before fitting their models.

A frozen comparison on preassigned PISA forms. The U.S. public-use student file of PISA 2012 contains 4,978 students in 162 schools [National Center for Education Statistics, 2014]. The survey randomly assigned three questionnaire forms, each containing a common section and two of three rotating question sets, to keep estimated response time at 30 minutes [OECD, 2014, Kastberg et al., 2014]. We use the common report of grade repetition in grades 1–6 as the binary target and three binary response indicators from each of the three rotated question sets. Each student has six of the nine selected predictors, and none has all nine. Asked-but-unanswered items count as nonendorsements, while items absent from a student's form remain unobserved. The 226 records without a valid common target response are excluded. Target nonresponse is balanced across forms: 72, 75 and 79 nonrespondents on forms A, B and C (Pearson $\chi^2_2 = 0.33$, $p = 0.85$); survey-weighted nonresponse rates are 4.4%, 5.0% and 5.2%, with replicate-weight z-statistics of at most 1.3 for the pairwise differences.

Forty schools train the old ensemble, another forty train the new one, and the remaining 82 supply 2,441 audit records. Each ensemble averages three block-specific forests, fitted using only students who received that block. The contrast depends on all nine predictors. Every pair of blocks occurs in a form, so both the accuracy and Brier differences decompose into observable functions (Appendix A24). Calibration uses incomplete training records only.

Table 1 compares optimized scores, a simple inverse-probability block/pair score, and twenty logistic chained-equation imputations with Rubin combination [van Buuren and Groothuis-Oudshoorn, 2011]. Survey-weighted intervals use the 80 supplied Fay replicates [Kastberg et al., 2014]. The audit domain is the set of students in the 82 audit schools; replicate estimates are computed for that domain with every replicate weight, the standard domain-estimation practice, and inference is conditional on the frozen models and the random school split. All three Brier intervals include zero, and the methods give similar estimates and widths. The randomized-accuracy gain is about 4.1 percentage points, a different target; both rules threshold to the negative class everywhere on this cube, so their deterministic accuracy difference is zero. The design interpretation assumes invariant reported answers across forms. Nine duplicated-item diagnostics find two unadjusted $p < 0.05$ differences (ST29Q02, 4.0 points between forms A and B; ST42Q03, 4.8 points between forms B and C), neither surviving Bonferroni correction. As a sensitivity analysis we equalize each item's weighted endorsement rate across its two forms by random flips in the audit sample, lowering the higher

Table 1: PISA secondary analysis on 2,441 held-out students. Brier improvement and 95% intervals are multiplied by 10^3; positive means the newer rule has smaller loss. Survey intervals use the 80 supplied Fay replicates. MI intervals also use Rubin combination under a working imputation model.

Method	Estimate	95% interval
Form score	1.13	$[-2.28, 4.53]$
Block/pair HT	1.24	$[-2.09, 4.57]$
Logistic MI	1.24	$[-2.12, 4.60]$

Table 2: Independent forests, cap 3: 60 binding tasks per dataset. Median fitted fixed/menu-branch variance ratio and oracle menu ratio (numerical lower endpoint). The branch menu includes every fixed triple and all 40 component trees. Neither column uses the unattainable full-forest target as its branching score.

Data	Fitted	Oracle menu	Fitted > 1.5
WDBC	1.085	1.090	0/60
Spambase	1.029	1.031	0/60
Ionosphere	1.117	1.100	0/60

form or raising the lower one, with 200 draws per scenario and the models and score tables held fixed. The mean shift of the optimized-score Brier estimate is at most 0.05 of its standard error, and the largest single shift is 0.35 standard errors. Equalizing both items at once shifts it by 0.015 standard errors on average. Complete-data truth and a branching counterfactual are unavailable, so this archive illustrates identification and uncertainty from real fixed forms, not an adaptivity ratio.

Independent forest updates with a restricted branching menu. On WDBC, Spambase and Ionosphere, twenty fixed splits each fit independent old and new random forests on disjoint training records [Wolberg et al., 1993, Hopkins et al., 1999, Sigillito et al., 1989]. Each forest has twenty depth-three trees. Nested sets of 3, 6, 9 and 12 training-binarized features give 240 tasks, all with cap $b = 3$. The fixed menu contains every feature triple. The branching menu of the earlier analysis contained those forms and all forty component trees, with jointly optimized leaf scores.

With this menu the fitted variance ratios have medians 1.03–1.12 and none exceeds 1.5 (Table 2, Figure 1). Appendix A20 replaces the restricted menu by all depth-three trees. The exact ratios have median 1.51 over the 180 binding tasks, and the restricted menu captures a median of 17% of the exact log-gain. Among 70 probability-output comparisons of the smaller independent-tree experiment, 60 already have active feature count above the cap, with population ratios of median 1.353 (SUSHI) and 1.295 (SAPA); for threshold outputs, 52 of 58 nonzero comparisons are nonbinding (Appendix S2.3).

Controlled gaps, transfer and operational costs. Depth-six CART/root refinements isolate a contrast that branching can compute in six queries. Exhaustive lower bounds over all fixed six-feature forms certify median population ratios of at least 1.883 on SUSHI and 1.474 on SAPA. The complete-calibration version is an efficiency diagnostic: its 500 independent target-population records already suffice for a direct audit at the chosen precision. Transferring calibration across SAPA age groups avoids that interpretation. Brier and clipped log-loss comparisons retain median fitted variance ratios 2.350 and 2.623, with conditional Wald coverage near 95%. A separate restricted-pilot experiment charges all newly collected calibration records (Appendices S6.1 and S3.2).

The imputation comparator in Tables 5 and 7 is a single deterministic regression imputation. Among the tree's path-set forms it selects the one whose calibration cell means best predict $Yd(X)$, asks that form of every audit record, and averages the calibration cell mean of $Yd(X)$ given the observed answers and label, with add-one shrinkage toward the overall calibration mean. It has no between-imputation variance component, so its intervals ignore calibration uncertainty, which explains its undercoverage. Proper multiple imputation is evaluated separately in Section 6 and Appendix A21.

If the cap can be relaxed, compare complete acquisition using its actual label cost. With fixed/complete

variance ratio $\rho > 1$, fixed form size b and complete size m, their continuous cost crossover is

$$c_Y^* = \frac{m - b\rho}{\rho - 1}. \tag{1}$$

Median crossovers on the controlled tasks are 3.16 and 3.74 feature-cost units. At $c_Y = 10$, complete records cost less on 45/45 SUSHI and 19/25 SAPA tasks (Figure 4). When the cap is immutable, complete acquisition is inadmissible. For nonlinear updates, the bias certificate must be checked first: WDBC GBDT/MLP budget curves reject overly tight caps regardless of sample size (Appendix S6.3).

S2 Independent Forest Updates and Binding Constraints

S2.1 Complete prespecified grid

The protocol fixes three datasets, twenty seeds 20261108–20261127, four nested feature counts $m \in \{3, 6, 9, 12\}$ and cap three. WDBC uses 100 old-training, 200 new-training and 269 evaluation records; Spambase uses 500, 1,000 and 3,101; Ionosphere uses 60, 120 and 171. We binarize strictly above pooled training medians. Among columns varying on these training records, take twelve evenly spaced column indices, then the first m. Feature selection does not inspect labels. Each version fits twenty depth-three trees with bootstrap samples, square-root feature sampling and minimum leaf size $\max(3, n_{\text{old}}/30)$ rounded down. The versions have separate training records and seeds. The target is $Y(p_1 - p_0)$ for probability averages; thresholding a forest may destroy degree-three identification.

Training records also fit the reference marginals, with uniform shrinkage 0.1 and a design floor $q_\iota \geq 0.05/|\mathcal{A}|$. These are records used to fit the evaluated models, so their in-sample paired loss is not an independent target audit. No fresh complete calibration sample is excluded from the acquisition accounting. Exact variances below condition on the fitted models, reference and empirical held-out law. Repeated splits reuse records, and their quantiles describe the experiment rather than independent population replications.

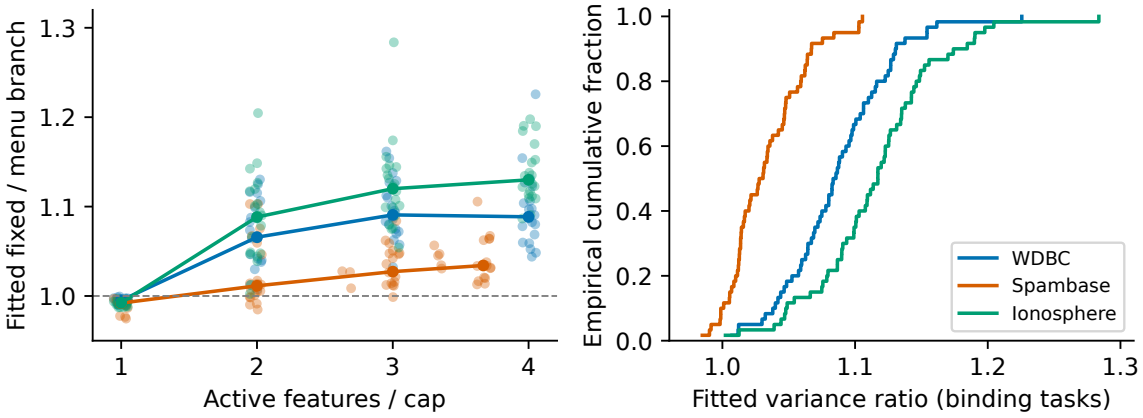


Figure 1: Every forest comparison and the distribution of the 180 binding cases. Lines in the left panel join dataset medians across candidate feature counts; models are refitted at each count, so the plot is not a causal or monotonicity claim about active features. The right panel shows the full empirical distribution.

S2.2 Branching denominator and numerical bounds

The fixed menu contains every size-three coordinate set. The tested branch menu contains those forms and all forty component trees of the two forests. Joint optimization uses the observed leaf, label and selected tree, with the same pointwise target identity as fixed-form optimization. It does not read all forest predictions on

one person. We also implement a centered random-component score: for signed forest weights w_j, reference centers μ_j, and $c = \sum_j w_j \mu_j$, return

$$H = Y\left(c + \frac{w_J\{f_J(X) - \mu_J\}}{q_J}\right).$$

Reference residual second moments choose q_J, with the same 0.05 uniform floor. The selected component's depth is at most three.

For each menu the reference-fitted procedure is evaluated on the held-out law. Its ratio can be below one: fitting error and the menu-specific positive probability floors can make the larger menu's implemented score worse. This is not a reversal of the optimal information comparison. Separate population-oracle programs omit the floor and jointly optimize the design and all cell scores. Their primal values and dual witnesses give numerical intervals, with explicit zero-mass-cell projections and full-cube unbiasedness repair. These are floating-point checks, not exact rational certificates. The maximum oracle ratio interval width is 0.00749; the maximum score or zero-cell residual is 2.3×10^{-15}.

Let W_F denote the unrestricted fixed-form optimum and W_M the optimum of the supplied branch menu. Since the menu is feasible, $W_{\text{branch}} \le W_M$, so W_F^L/W_M^U lower-bounds W_F/W_{branch}. In contrast W_F^U/W_M^L is only an upper endpoint for W_F/W_M. The full-information variance supplies a separate, typically looser upper bound $W_F^U/\operatorname{Var}(Yd)$ for the unrestricted ratio. We do not infer that every depth-three adaptive strategy is close to the fixed optimum. None of the tested fitted ratios exceeds 1.5; their maximum is 1.284. The largest oracle-menu lower endpoint is 1.207. Appendix A20 computes W_{branch} over all depth-three trees by certified column generation, on a full-support version of the held-out law. The exact ratios have median 1.51 on the binding tasks, so this supplied menu understates the intrinsic ratio substantially.

Table 3: Full independent-forest grid: each row contains 20 fixed splits. Fitted ratio quantiles describe variation across overlapping splits; they are not confidence intervals. Oracle intervals concern the specified branch menu.

Data	m	Active median	Fitted median	Fitted 10–90%	Oracle lower	Oracle upper
Ionosphere	3	3	0.992	[0.987, 0.995]	1.0000	1.0000
Ionosphere	6	6	1.088	[1.036, 1.143]	1.0699	1.0707
Ionosphere	9	9	1.120	[1.076, 1.158]	1.0992	1.0994
Ionosphere	12	12	1.130	[1.107, 1.190]	1.1053	1.1058
Spambase	3	3	0.992	[0.978, 0.995]	1.0000	1.0000
Spambase	6	6	1.011	[0.991, 1.078]	1.0258	1.0258
Spambase	9	9	1.027	[1.011, 1.059]	1.0332	1.0335
Spambase	12	11	1.034	[1.016, 1.064]	1.0362	1.0364
WDBC	3	3	0.995	[0.991, 0.998]	1.0000	1.0000
WDBC	6	6	1.066	[1.012, 1.117]	1.0861	1.0878
WDBC	9	9	1.091	[1.062, 1.139]	1.0941	1.0976
WDBC	12	12	1.089	[1.052, 1.133]	1.0810	1.0845

S2.3 Structural ties in the independent-tree benchmark

The eight-feature experiment in Appendix S3.1 must be stratified per task, rather than comparing separate medians of active feature count and cap. Among probability-output contrasts, 39/45 SUSHI and 21/25 SAPA tasks are binding. Within those strata, median population ratios are 1.353 and 1.295; median global ratios remain 1.000. The other ten tasks have a form that computes their contrast exactly. Among 58 nonzero thresholded contrasts, 35/39 SUSHI and 17/19 SAPA tasks are nonbinding. The six binding threshold tasks have median population ratios 1.289 and 1.360 in the two datasets. Thus nonbinding budgets explain most thresholded ties, not the probability-output results. The released `independent_binding_strata.csv` contains every stratum.

Table 4: All eight-feature comparisons, including ranking-feasible profiles and the full-support population $P_{.01}$. Active counts coordinates on which the contrast depends. Twelve cube-domain deterministic updates are identically zero and omitted. Numerical certificate endpoints are released per task.

Data / rules / domain	Tasks	Med. b	Active	ρ_{wc}	ρ_P	$\rho_{P_{.01}}$	Max. $\rho_{P_{.01}}$
SUSHI / prob. / cube	45	4	6	1.000	1.301	1.301	1.921
SUSHI / prob. / ranking	45	4	6	1.000	1.301	1.300	1.922
SUSHI / det. / cube	39	4	4	1.000	1.000	1.000	1.432
SUSHI / det. / ranking	39	4	4	1.000	1.000	1.000	1.432
SAPA / prob. / cube	25	4	6	1.000	1.260	1.262	2.435
SAPA / det. / cube	19	4	3	1.000	1.000	1.000	1.489

Table 5: Primary-split larger-tree comparisons, conditional on 500 complete calibration records. Median variance ratios for unbiased scores; imputation uses MSE at $n = 1000$. Oracle entries use upper certificates on the supplied menu, not all forms. Ordinal form scores use tree-threshold categories. Each task uses the fixed primary split; repeated-split diagnostics are reported separately.

Data	b	Tasks	Prefix	Form scores	Oracle lower	Oracle upper	Imputation
SUSHI	3	45	2.613	2.607	2.532	2.532	0.948
SUSHI	6	45	3.765	3.403	3.184	3.184	0.835
SAPA	3	25	2.453	2.303	2.255	2.255	1.031
SAPA	6	25	2.758	2.438	2.286	2.286	1.030
SAPA-ordinal	3	25	2.393	2.232	2.162	2.162	1.388
SAPA-ordinal	6	25	3.088	2.778	2.389	2.389	1.235

S3 Complete Form Scores: Earlier Operational Comparisons

S3.1 Independent Model Versions on Eight Features

For each of the 45 SUSHI and 25 binary SAPA primary-split tasks, we select eight target-excluding features at equally spaced indices of the existing feature list. A depth-two old classifier is fitted on the first 400 records; a depth-three new classifier is independently fitted on the next 400. Each uses minimum leaf size 20 and a fixed seed. The remaining records define the evaluation law. We examine both probability outputs and deterministic threshold outputs.

The audit first follows the old tree, then the new tree, reusing an answer when both trees ask the same feature. Its cap is the maximum size of the union of the two reached paths over all 256 profiles, at most five. This cap is computed from the frozen rules, not chosen using the evaluation population. Hence the branching coefficient is $\|d\|_\infty^2$ and its population variance is $\mathrm{Var}_P(Yd)$. We optimize all fixed forms at that same cap, including every possible answer table. Both coefficients and population-specific variances have independently evaluated primal/dual bounds.

For SUSHI, we additionally enumerate all 8! orders of the eight nontarget alternatives and retain only feasible profiles of the eight comparisons. The optimization is repeated on this ranking support, while retaining the same audit cap. This checks the effect of logical restrictions among pairwise comparisons. The target pair involves the two other alternatives, so either target label is compatible with each retained order. No individual-level source records are released.

Besides the empirical population P, we report $P_{.01} = 0.99P + 0.01\mathsf{U}$, where U is uniform on the stated profile–label space. This fixed smoothing has full support and gives a direct example covered by Proposition 15. It is a diagnostic regularization of the population, not an estimate that the data-generating law has that contamination fraction.

S3.2 EM Calibration from Restricted Pilot Records

For uniformly assigned forms, record only the form, stopping-prefix cell, and label. Let $N_{\iota,c,y}$ be the resulting counts, and let $c_\iota(\ell)$ be the observed prefix for leaf ℓ. The joint leaf–label probabilities $p_{\ell y}$ maximize the

penalized observed-data log likelihood

$$\sum_{\iota,c,y} N_{\iota,c,y} \log\left(\sum_{\ell:c_\iota(\ell)=c} p_{\ell y}\right) + \tfrac{1}{2}\sum_{\ell,y} \log p_{\ell y}, \qquad p \geq 0, \quad \sum_{\ell,y} p_{\ell y} = 1.$$

Form-assignment probabilities are known constants in this likelihood. The half-count penalty prevents zero reference masses. The EM update assigns each cell count among its compatible leaves in proportion to their current probabilities, adds half a count to each leaf–label state, and renormalizes. It uses unmatched records as coarse observations; it is not allowed to use their hidden leaves. The implementation stops at maximum coordinate change below 10^{-9}, or 2,000 iterations, and reports the achieved change and iteration count. A separate constrained-likelihood optimizer checks the update on a small example.

The experiment freezes the 70 primary-split depth-six rules. Its population is the empirical law of all records after the 800 training records, rather than the held-out law of the conditional-calibration experiment. Each task has 500 iid simulated pilots, at each of $n_0 = 100, 250, 500$ respondents. Pilot sizes share the beginning of the same random sequence to improve paired comparisons. Every pilot respondent obeys the six-feature cap, and every pilot record counts toward $N \in \{1000, 2000, 5000\}$.

We compare matched-path reference counts with EM reference estimates, fitting the same prefix-score program afterwards. We also separate the reference estimate from the score used to reuse the pilot. One pilot score is completion-only weighting; a second is a globally unbiased prefix score with uniform form probabilities, fixed in advance from the 800 historical training records. The latter uses incomplete paths without fitting an estimator to the same pilot that is being averaged.

For pilot mean T_0 and second-stage score mean T_1, use $\lambda T_0 + (1-\lambda)T_1$. We compare $\lambda = 0$, $\lambda = n_0/N$, and

$$\lambda_{\text{prior}} = \frac{\widehat{V}_1/(N-n_0)}{\widehat{V}_0/n_0 + \widehat{V}_1/(N-n_0)},$$

where both variance predictions come exclusively from the fixed historical reference, before sampling the pilot. These weights are not selected with evaluation data or with the realized pilot. Conditional on a pilot, the exact risk is

$$\lambda^2(T_0 - \Delta)^2 + \frac{(1-\lambda)^2}{N-n_0}\operatorname{Var}_P(H_{\text{fitted}}).$$

The second-stage score remains unbiased for every fitted reference, so no cross term remains. Averaging this expression over pilots yields the reported risk estimates. Per-task Monte Carlo standard errors accompany each comparison. These are risk estimates conditional on the frozen model and empirical population, not independent replications of a deployment study. All newly acquired records are capped at six features; with unit feature and label costs the same comparison spends $7N$ total answers.

At $N = 1000, n_0 = 500$, changing only matched counts to EM while retaining completion-weighted count-proportional reuse changes the median risk ratio from 8.97 to 8.15 on SUSHI and from 7.30 to 7.02 on SAPA. Combining EM with the stable historical pilot score and prior weight gives 4.50 and 3.05. The associated median per-task Monte Carlo standard errors are 0.121 and 0.083. At $n_0 = 100$ this last protocol gives 4.08 and 2.97, with median errors 0.021 and 0.014. These paired comparisons identify the noisy pilot score as a separate cost from reference estimation. Across all 105,000 fits, EM reached the specified coordinate tolerance in at most 629 iterations; none hit the 2,000-iteration cap. No claim of global optimality over pilot sizes or allocation policies follows.

S3.3 Intervals for Frozen Unbiased Scores

With a fixed design fitted on separate historical or calibration records, the audit observations $H_1, \ldots, H_n$ are iid conditional on that fitting data and have mean Δ. Use the ordinary sample variance $s_H^2 = (n-1)^{-1}\sum_i (H_i - \overline{H})^2$ and the Wald interval $\overline{H} \pm 1.96 s_H/\sqrt{n}$. For a finite menu with positive used probabilities and positive variance, its conditional coverage tends to 0.95. This is an asymptotic statement; a small audit can miss rare large importance scores. A finite-sample alternative uses bounded-score concentration, at the cost of wider intervals.

Table 6: Pilot-inclusive risk relative to branching, median across tasks. Each cell integrates the second stage exactly and averages 500 pilot replications per task. All new people obey the six-feature cap. Matched and EM estimate the reference; HT and stable specify the prespecified pilot score. Count, prior and discard specify reuse weights. The last column is the median per-task Monte Carlo standard error for EM/stable/prior, not uncertainty across populations.

Data	N	n_0	Matched/HT	EM/HT	EM/stable	EM/prior	EM/discard	MC SE
SUSHI	1000	100	5.71	4.81	4.09	4.08	4.44	0.021
SUSHI	1000	250	7.32	6.04	4.30	4.21	5.37	0.054
SUSHI	1000	500	8.97	8.15	4.61	4.50	8.01	0.121
SUSHI	2000	100	5.33	4.37	4.05	4.02	4.21	0.012
SUSHI	2000	250	6.42	4.98	4.15	4.13	4.60	0.026
SUSHI	2000	500	6.97	6.06	4.29	4.25	5.34	0.054
SUSHI	5000	100	5.10	4.12	4.01	4.01	4.08	0.007
SUSHI	5000	250	5.93	4.38	4.07	4.06	4.24	0.012
SUSHI	5000	500	5.87	4.79	4.13	4.07	4.45	0.021
SAPA	1000	100	4.24	3.69	2.99	2.97	3.22	0.014
SAPA	1000	250	5.81	4.97	3.07	3.01	3.87	0.038
SAPA	1000	500	7.30	7.02	3.09	3.05	5.74	0.083
SAPA	2000	100	3.91	3.32	2.94	2.93	3.05	0.008
SAPA	2000	250	4.78	3.92	2.94	2.94	3.31	0.019
SAPA	2000	500	5.62	4.95	2.93	2.92	3.83	0.036
SAPA	5000	100	3.69	3.10	2.91	2.91	2.95	0.005
SAPA	5000	250	4.24	3.32	2.92	2.92	3.05	0.008
SAPA	5000	500	4.46	3.69	2.88	2.89	3.19	0.014

We evaluate conditional coverage on the first five primary-split tasks in each of SUSHI, binary SAPA and ordinal SAPA. The selection is by task order, not observed coverage. Models and the complete 500-record calibration are frozen. For each of $n = 100, 1000$, 2,000 iid empirical-population audits compare branching, prefix scores, calibrated full-form scores, and the fitted imputation comparator. Coverage is against the true gain of that empirical population. Simulation standard errors are reported. These results do not validate a Gaussian interval for an unknown imputation bias: its interval generally targets the imputed mean instead of Δ. Nor do these single-stage intervals automatically cover an estimator that reuses an adaptive pilot. With a prespecified pilot weight, the two-stage variance must include both $\lambda^2 s_0^2/r$ and $(1-\lambda)^2 s_1^2/(N-r)$; further conditions or a sequential interval are needed to justify its normal approximation.

S3.4 Stratified Validation under Expected Cost

To strengthen the relaxed-budget comparison, we also implement a two-phase augmentation related to known-coarsening estimators and prediction-powered inference [Robins et al., 1994, Tao et al., 2020, Angelopoulos et al., 2024]. First observe a small fixed form and the label, denoted U. Independently conditional on U, acquire the remaining features with probability $\pi(U) > 0$. For any frozen prediction $m(U)$, the score

$$H = m(U) + \frac{R}{\pi(U)}\{Yd(X) - m(U)\}$$

is conditionally unbiased given (X, Y), and

$$\mathrm{Var}(H) = \mathrm{Var}(Yd) + \mathbb{E}\big[(\pi(U)^{-1} - 1)(Yd - m(U))^2\big] .$$

For a fixed expected validation fraction, minimizing the second term uses $\pi(u)$ proportional to the square root of the conditional squared residual, clipped to the permitted probability range. This is the usual Neyman allocation principle applied to augmentation residuals.

The same 500 complete calibration records fit smoothed cell means and residual moments. Candidate initial forms are empty, the root alone, or the root plus one other feature from a path form. Candidate mean validation fractions are $0.05, 0.1, 0.2, 0.4, 0.7, 1$, with $\pi \geq 0.02$. The initial form and fraction minimize

Table 7: Conditional coverage of nominal 95% Wald intervals over the first five tasks per dataset, selected before inspecting coverage. Each task uses 2,000 iid audit replications with frozen training and calibration. Mean/min/max are descriptive summaries; the final column is the largest binomial Monte Carlo standard error within that row. Imputation intervals need not cover the target gain.

Data	n	Score	Mean	Min.	Max.	MC SE
SUSHI	100	Branching	0.944	0.939	0.949	0.005
SUSHI	1000	Branching	0.948	0.944	0.949	0.005
SUSHI	100	Prefix	0.947	0.942	0.954	0.005
SUSHI	1000	Prefix	0.948	0.945	0.955	0.005
SUSHI	100	Form scores	0.949	0.944	0.954	0.005
SUSHI	1000	Form scores	0.950	0.941	0.956	0.005
SUSHI	100	Imputation	0.934	0.879	0.955	0.007
SUSHI	1000	Imputation	0.797	0.242	0.951	0.010
SAPA	100	Branching	0.947	0.941	0.953	0.005
SAPA	1000	Branching	0.948	0.941	0.955	0.005
SAPA	100	Prefix	0.952	0.947	0.957	0.005
SAPA	1000	Prefix	0.949	0.945	0.953	0.005
SAPA	100	Form scores	0.942	0.940	0.948	0.005
SAPA	1000	Form scores	0.954	0.951	0.957	0.005
SAPA	100	Imputation	0.938	0.919	0.949	0.006
SAPA	1000	Imputation	0.782	0.618	0.946	0.011
SAPA-ordinal	100	Branching	0.944	0.939	0.948	0.005
SAPA-ordinal	1000	Branching	0.948	0.944	0.958	0.005
SAPA-ordinal	100	Prefix	0.945	0.942	0.950	0.005
SAPA-ordinal	1000	Prefix	0.951	0.949	0.958	0.005
SAPA-ordinal	100	Form scores	0.951	0.948	0.957	0.005
SAPA-ordinal	1000	Form scores	0.949	0.944	0.953	0.005
SAPA-ordinal	100	Imputation	0.265	0.018	0.503	0.011
SAPA-ordinal	1000	Imputation	0.000	0.000	0.001	0.001

estimated variance times estimated cost using calibration alone. We then evaluate exact variance and expected cost on the held-out population. Illustrative label costs are $c_Y = 1, 10, 100$ feature-answer units; these are sensitivity values, not measured medical prices. Branching is charged for its actual mean path length, not unused padding questions. The reported coefficient ratio is

$$\frac{[c_Y + |T| + (m - |T|)\mathbb{E}\pi(U)]\,\mathrm{Var}(H)}{[c_Y + \mathbb{E}\,\mathrm{depth}(X)]\,\mathrm{Var}(Yd)}.$$

This is a large-budget, expected-cost comparison conditional on available calibration. The validation decision depends on observed answers and can request a complete profile, so it belongs to a relaxed two-phase workflow, not to the hard-cap fixed-form class. It does not enforce an exact realized total answer budget. It is reported separately from the finite total-budget experiments to avoid conflating these constraints.

S4 Certificates over All Forms and Precision-Cost Comparisons

S4.1 A Population Witness that Checks Every Fixed Form

For a centered function $h(X, Y)$ under P, put

$$I_T(h) = \mathbb{E}_P\big[\mathbb{E}_P[h \mid X_T, Y]^2\big], \qquad \mathcal{L}_P(h) = \frac{\mathbb{E}_P[(t - \Delta)h]^2}{\max_{|T|=b} I_T(h)}.$$

For any globally pointwise-unbiased score H, with form chosen before this person's answers, conditioning and Cauchy–Schwarz give

$$\mathbb{E}_P[(t - \Delta)h] = \mathbb{E}_{P,q}[(H - \Delta)\mathbb{E}(h \mid X_T, Y)], \qquad \mathrm{Var}_{P,q}(H) \geq \mathcal{L}_P(h).$$

Table 8: Expected-cost coefficient ratios for stratified validation and complete-record averaging, relative to branching charged by actual mean path length. Calibration is historical and held fixed. Label cost c_Y is in feature-answer units. Validation can request all features and depends on initial answers; it is outside the hard-cap fixed-form class.

Data	c_Y	Tasks	Stratified validation	Complete mean	Validation fraction
SUSHI	1	45	4.690	4.966	0.700
SUSHI	10	45	2.556	2.561	1.000
SUSHI	100	45	1.221	1.221	1.000
SAPA	1	25	3.167	3.572	0.688
SAPA	10	25	1.999	2.016	0.721
SAPA	100	25	1.144	1.144	1.000
SAPA-ordinal	1	25	3.683	3.815	0.397
SAPA-ordinal	10	25	2.081	2.068	0.691
SAPA-ordinal	100	25	1.148	1.148	1.000

Consequently $W_{\text{fixed}}(P) \geq \mathcal{L}_P(h)$. By Proposition 2, $\mathcal{L}_P(h)/n$ also bounds globally unbiased sequential estimators. The one-dimensional submodel with density $1 + \theta h$ on the support of P gives the corresponding local asymptotic lower bound for arbitrary estimators. It does not give a pointwise finite-sample lower bound for biased procedures. Since conditional projection norms increase on refinement, checking all size-b forms also checks every smaller form.

For the empirical law of N_e records, a form's information is

$$I_T(h) = \frac{1}{N_e} \sum_{c:N_{T,c}>0} \frac{(\sum_{i:O_{T,i}=c} h_i)^2}{N_{T,c}}, \qquad O_{T,i} = (X_{i,T}, Y_i).$$

We optimize witnesses constant on a tree-leaf/label cell, starting from the path-form menu and adding up to five most-violated forms per iteration. The separation step exhaustively evaluates all $\binom{m}{b}$ forms. A restricted witness space can weaken the lower bound but cannot turn it into a menu-restricted lower bound. We retain the best certified witness, irrespective of whether optimization converges. Scores are quantized to 2^{-28}, recentered, and re-evaluated; cell sums use extended precision. The reported bound subtracts 10^{-12} from the absolute covariance and adds 10^{-10} to the maximum information before division. These are numerical certificates; the separate four-bit product certificate below uses exact rational arithmetic.

The full primary-split census consists of all 45 SUSHI targets and all 25 binary SAPA targets. Each tree has depth six, 800 training and 500 historical calibration records; the remaining records define P. Thus targets and populations match the larger-table primary split. Every task checks 376,740 forms for SUSHI or 38,760 for SAPA. The initial ten-task subset was chosen by task order to keep exhaustive computation manageable; it contained only SAPA's A-domain targets and was not representative of all domains. The complete census removes that task-selection limitation within this benchmark. The underlying dataset/model-family limitation remains. No full 2^m cube is needed for the population witness.

The released task rows give each bound and a maximizing form, while the companion witness file permits re-evaluation. The new tasks are also checked on the maximizing form and 100 randomly selected forms by an independent NumPy grouping calculation. Table 9 separates each SAPA domain and the ten-task diagnostic subset from the remaining tasks.

S4.2 Respondent Counts for a Target Interval Width

For the first five tasks in each binary dataset, consider target half-widths $e = 0.01, 0.025$. For each unbiased score of variance v we use the planning count

$$n_e = \max\{30, \lceil 1.9599639845^2 v/e^2 \rceil\}.$$

The variance here is the exact evaluation-population variance, so this is an oracle benchmark planning curve, not a sample-size prescription learned from 500 calibration records. At that count, 2,000 new iid audit

Table 9: All eligible tasks, with domain and ten-task diagnostic summaries. Task rows and maximizing forms are supplied in all_form_bounds_all.csv; the median endpoints do not describe a single common task.

Tasks	N	Min. LB	Med. LB	Max. LB	Med. UB
SUSHI, all	45	1.433	1.883	2.362	3.184
SAPA, all	25	1.179	1.474	2.167	2.286
SAPA, A	5	1.236	1.474	1.801	2.169
SAPA, C	5	1.402	1.593	1.625	2.319
SAPA, E	5	1.357	1.469	1.634	2.130
SAPA, N	5	1.179	1.280	1.425	1.737
SAPA, O	5	1.577	1.795	2.167	3.130
SUSHI, first five	5	1.775	1.853	2.067	3.184
SUSHI, remaining tasks	40	1.433	1.899	2.362	3.207
SAPA, first five	5	1.236	1.474	1.801	2.169
SAPA, remaining tasks	20	1.179	1.523	2.167	2.302

replications check ordinary studentized Wald coverage and mean half-width. Both the design and score were fitted before the evaluation population was used. Counts are not tuned by searching the coverage simulations. We report coverage rather than promise that normal approximations guarantee 95% at every finite sample size.

Branching, prefix and complete-form scores obey the hard cap. A fourth method uses the stratified proxy correction from Appendix S3.4:

$$H_{\mathrm{val}} = c(U) + \frac{R}{\pi(U)}\{t - c(U)\}.$$

It may acquire all features and belongs to the relaxed-cap comparison. Its calibration-only proxy and validation probability are chosen separately at label costs $c_Y = 1, 100$. We integrate its two possible score values per record with probabilities $1 - \pi(U)$ and $\pi(U)$. Costs charge initial features, expected validation features and every label. Branching is charged by its actual path length. All calibration records are historical in this experiment; fresh pilot costs are analyzed separately in Appendix S3.2.

S4.3 Probability-Floor Sensitivity

We repeat complete-form fitting at $\beta = 0.01, 0.05, 0.1, 0.2$ on the same ten tasks. At each floor, the existing 350/150 internal split selects uniform shrinkage from $0.02, 0.1, 0.3, 1$, then the score is refitted on all 500 calibration records. This comparison holds the full-form score family fixed; it does not add the optional prefix fallback. Evaluation records choose neither floor nor shrinkage. The medians range from 3.302 to 3.374 on SUSHI and 2.420 to 2.436 on SAPA. The released task rows include probabilities and selected shrinkage; these ten tasks do not establish universal insensitivity to the floor.

S5 A Product Counterexample with an Exact Rational Certificate

Let f be the four-feature contrast in Proposition 16 and $\alpha = (7 + 2\sqrt{2})/4$. On two disjoint blocks define $F(x, z) = f(x)f(z)$. Then

$$V_{\mathrm{branch}}(F, 6) = 1, \qquad V_{\mathrm{fixed}}(F, 6) < \frac{541}{100} < \alpha^2.$$

In particular the fixed coefficient is not generally multiplicative when forms can reallocate queries between blocks. The exact product-address result requires its separate information-per-cost argument.

Here is a finite rational certificate for the strict upper bound. The supplement's `product_certificate.csv` supplies positive integers Q_T, their sum Q_Σ, and integers $G_{T,c}$ for all 28 six-feature forms and 64 cells per form. Set

$$q_T = Q_T/Q_\Sigma, \qquad g_{T,c} = G_{T,c}/2^{28}.$$

Table 10: Precision planning on five prespecified tasks per dataset. Counts and costs are medians; coverage and half-width are Monte Carlo means; minimum is across the five task coverage rates. Each rate uses 2,000 replications. †Validation belongs to the relaxed-cap class.

Data	e	c_Y	Method	n_e	Cost	Cov.	Min.	Half-width
SUSHI	0.010	1	Branching	834	4906	0.952	0.948	0.0100
SUSHI	0.025	1	Branching	134	788	0.947	0.941	0.0249
SUSHI	0.010	1	Prefix	3299	23093	0.952	0.949	0.0100
SUSHI	0.025	1	Prefix	528	3696	0.949	0.946	0.0249
SUSHI	0.010	1	Form scores	2837	19859	0.950	0.943	0.0100
SUSHI	0.025	1	Form scores	454	3178	0.950	0.939	0.0249
SUSHI	0.010	1	Validation†	1094	23194	0.950	0.943	0.0100
SUSHI	0.025	1	Validation†	176	3731	0.948	0.942	0.0249
SUSHI	0.010	100	Branching	834	87472	0.950	0.946	0.0100
SUSHI	0.025	100	Branching	134	14054	0.952	0.946	0.0249
SUSHI	0.010	100	Prefix	3299	349694	0.948	0.943	0.0100
SUSHI	0.025	100	Prefix	528	55968	0.951	0.947	0.0249
SUSHI	0.010	100	Form scores	2837	300722	0.952	0.949	0.0100
SUSHI	0.025	100	Form scores	454	48124	0.951	0.945	0.0250
SUSHI	0.010	100	Validation†	834	106752	0.953	0.950	0.0100
SUSHI	0.025	100	Validation†	134	17152	0.947	0.939	0.0249
SAPA	0.010	1	Branching	648	3668	0.946	0.943	0.0100
SAPA	0.025	1	Branching	104	589	0.945	0.931	0.0248
SAPA	0.010	1	Prefix	1536	10752	0.950	0.947	0.0100
SAPA	0.025	1	Prefix	246	1722	0.952	0.947	0.0249
SAPA	0.010	1	Form scores	1233	8631	0.949	0.939	0.0100
SAPA	0.025	1	Form scores	198	1386	0.951	0.948	0.0249
SAPA	0.010	1	Validation†	855	12797	0.951	0.943	0.0100
SAPA	0.025	1	Validation†	137	2048	0.947	0.945	0.0249
SAPA	0.010	100	Branching	648	67820	0.950	0.943	0.0100
SAPA	0.025	100	Branching	104	10885	0.949	0.946	0.0249
SAPA	0.010	100	Prefix	1536	162816	0.949	0.945	0.0100
SAPA	0.025	100	Prefix	246	26076	0.952	0.947	0.0249
SAPA	0.010	100	Form scores	1233	130698	0.949	0.942	0.0100
SAPA	0.025	100	Form scores	198	20988	0.948	0.942	0.0249
SAPA	0.010	100	Validation†	648	77760	0.955	0.949	0.0100
SAPA	0.025	100	Validation†	104	12480	0.948	0.945	0.0249

On each of the 256 sign profiles, the supplied exact-arithmetic checker verifies

$$\sum_T G_{T,c_T(x,z)} = 2^{28}F(x,z), \qquad \sum_T \frac{Q_\Sigma G^2_{T,c_T(x,z)}}{2^{56}Q_T} < 541/100.$$

These finite integer/rational inequalities directly imply feasibility and the stated uniform second-moment upper bound; a solver status or floating-point objective is not used as the proof. The certificate includes forms allocating two and four coordinates to the two blocks, as well as three and three. The checker can verify the released certificate without calling an optimizer. For context, the separate numerical primal/dual bracket is $[5.40562454, 5.40562463]$, whereas $\alpha^2 = 6.03737373\ldots$; the exact argument only asserts the displayed strict upper bound.

S6 Calibration Transport, Cost Crossovers, and Budget Curves

S6.1 Cross-Age Calibration and Proper Scores

The protocol is fixed in `PROTOCOL_v10.json`. Each of the 25 SAPA targets uses the same twenty target-excluding features as the main benchmark. We apply permutation seed 20261106 before dropping records

Table 11: Calibration in both directions. Ratios are medians; coverage averages 1,000 simulated audits of size 1,000 per case. Source discrepancy is the median absolute difference of reference and evaluation means. Misses count source-reference intervals that omit the empirical target, a descriptive transport diagnostic.

Metric	Direction	Cases	Matched	Shifted	Coverage	Discrepancy	Misses
Randomized accuracy	$\leq 25 \to > 25$	25	2.542	2.548	0.950	0.0053	1/25
Randomized accuracy	$> 25 \to \leq 25$	25	2.455	2.518	0.947	0.0082	8/25
Threshold accuracy	$\leq 25 \to > 25$	23	1.049	1.056	0.949	0.0206	5/23
Threshold accuracy	$> 25 \to \leq 25$	23	1.054	1.055	0.950	0.0220	10/23
Brier improvement	$\leq 25 \to > 25$	25	2.372	2.351	0.951	0.0029	4/25
Brier improvement	$> 25 \to \leq 25$	25	2.237	2.349	0.950	0.0049	5/25
Clipped log loss	$\leq 25 \to > 25$	25	2.596	2.555	0.949	0.0143	3/25
Clipped log loss	$> 25 \to \leq 25$	25	2.596	2.641	0.948	0.0181	4/25

missing a selected feature, target, or age. The first 800 records fit the depth-six tree with minimum leaf size 25 and tree seed 623. They belong to neither calibration nor evaluation. Age at most 25 defines the younger group and age above 25 the older group. Each group's first 500 remaining records form its reference sample; its remaining 305–370 records define the empirical evaluation law. Both directions are reported for every task. The matched reference is an explicitly additional diagnostic resource, not an assumed input to the source-reference audit. Conditional on the frozen model, target-representative complete records permit direct estimation whenever they are available.

We compare probability-matching accuracy, threshold accuracy, Brier improvement and log-loss improvement clipped at 0.01 and 0.99. The loss tables use the same frozen root and leaf probabilities in every condition. Each target is constant on the same tree leaves, so branching calculates it using at most six features. The form program uses all observed form coordinates and separate constraints for each label. Its marginal reference tables mix the 500-record frequencies with 0.1 times the uniform law on each form–label space. This shrinkage and the probability floor $q_T \geq 0.1/L$ are fixed before evaluation, without target-population tuning. No whole-profile enumeration is needed for this fit.

For every source-fitted and matched-fitted score we compute conditional means and variances by exact finite sums over target records and random form assignment. We also draw 1,000 independent audits of size 1,000 and evaluate ordinary sample-variance Wald intervals. Numerical mean errors and algebraic constraint residuals are released. Coverage summaries average conditional simulations, with per-task Monte Carlo errors supplied; they do not estimate deployment-level uncertainty from 50 independent cohorts. Two thresholded tasks have identically zero contrasts in both directions and are reported separately. All other task/metric/direction combinations are included irrespective of the observed efficiency ratio.

The source-reference direct mean is also reported as a transport diagnostic. Its discrepancy from the target mean combines group differences and finite reference/evaluation samples. Counting whether its conventional 500-record interval contains the empirical target is descriptive; it is not a repeated-sampling coverage estimate for a population-level age effect. Pointwise-unbiased form scores use source data only to choose variance, and target audit records to estimate the target mean.

S6.2 Complete Records and Intermediate Label Cost

For fixed fitted scores and a common target population, let $\sigma_F^2 = \rho\sigma_B^2$ denote fixed and branching variances. Complete records compute the branching target directly and have variance σ_B^2. Ignoring the integer rounding of a planned sample size, the ratio of total costs for equal variance is

$$\frac{C_F}{C_{\text{complete}}} = \rho\frac{c_Y + b}{c_Y + m}.$$

Solving its equality with one yields (1). If $\rho = 1$ and $b < m$, the fixed design remains cheaper at every finite label cost; if $m - b\rho \leq 0$, complete acquisition is weakly cheaper for all nonnegative label costs. The released rows also use exact integer planning counts n_B, n_F and their crossover $(mn_B - bn_F)/(n_F - n_B)$ whenever $n_F > n_B$. The same-population complete-reference half-width is $1.95996398454\sqrt{\sigma_B^2/500}$.

Table 12: All-task planned fixed-form/complete-record cost ratios at half-width 0.025. Counts include integer rounding and a minimum of 30. The last column counts tasks on which complete acquisition is cheaper. Both designs condition on supplied calibration.

Data	c_Y	Tasks	Cost ratio	Complete cheaper
SUSHI	0	45	0.726	1/45
SUSHI	1	45	0.818	5/45
SUSHI	3	45	0.984	17/45
SUSHI	10	45	1.427	45/45
SUSHI	30	45	2.103	45/45
SUSHI	100	45	2.806	45/45
SAPA	0	25	0.731	3/25
SAPA	1	25	0.812	5/25
SAPA	3	25	0.953	11/25
SAPA	10	25	1.299	19/25
SAPA	30	25	1.754	25/25
SAPA	100	25	2.152	25/25

Table 13: Intermediate-cost diagnostic at $c_Y = 10$ on the ten fixed tasks. Validation is refitted at that cost, with 2,000 conditional coverage simulations per task. Other rows reuse verified variances.

Data	Method	Records	Cost	Validation coverage
SAPA	branching	104	1525	–
SAPA	complete	104	3120	–
SAPA	debiased proxy	108	3120	0.946
SAPA	full form	198	3168	–
SAPA	prefix	246	3936	–
SUSHI	branching	134	1994	–
SUSHI	complete	134	5092	–
SUSHI	debiased proxy	134	5092	0.948
SUSHI	full form	454	7264	–
SUSHI	prefix	528	8448	–

Complete acquisition is admissible only in the relaxed-cap comparison. Randomly validating a subset of a larger enrolled cohort incurs a different cost if labels are obtained on every enrollee. The residual-correction comparator charges that cost explicitly; direct full-record sampling enrols only the records it actually observes. This distinction is necessary when label costs are substantial.

The all-task cost and precision calculations use all 70 existing frozen score variances. Branching path lengths are recomputed from the same frozen training protocol, with exact target-variance agreement checked. Label costs are $0, 1, 3, 10, 30, 100$. The ten-task correction diagnostic reoptimizes its source-calibrated candidate forms, validation fractions and within-form strata at $c_Y = 10$, using the same candidate grid as Appendix S3.4. Its new conditional coverage simulation has 2,000 audits per task. No calibration or pilot cost is silently included in the per-record costs: the displayed comparisons condition on a supplied reference, whereas Appendix S3.2 charges fresh capped calibration.

S6.3 Identification Floors Across the Whole Budget Range

We reuse the twenty binary12 prediction grids from Appendix A27.1, with no refitting or seed selection. For each cap from zero to twelve, we seek feasible primal and dual certificates for $\delta_b = \inf_{\deg(v)\le b} \|d - v\|_\infty$. The target comparison is with the fixed tolerance 0.025. Let F_b contain the degree-at-most-b Walsh columns. When the complementary set is smaller, an equivalent residual program minimizes $\|r\|_\infty$ subject to $F_{>b}^{\mathsf{T}} r = F_{>b}^{\mathsf{T}} d$; then $d - r$ has degree at most b. The code chooses the smaller coefficient representation.

Every LP has a 45-second solver limit. A feasible Fourier truncation and its high-degree residual supply bounds if the optimizer does not finish. Primal coefficients are rounded to multiples of 2^{-36}. Rounded dual

Table 14: Certified bracket for the smallest cap with $\delta_b < 0.025$. A lower endpoint l means every cap below l is certified insufficient; the upper endpoint has a feasible approximation below the tolerance. Equality identifies this minimum exactly.

Model	Output	Seed	Minimum cap LB	Minimum cap UB
GBDT	probability	20261108	4	5
MLP	probability	20261108	7	9
GBDT	threshold	20261108	6	6
MLP	threshold	20261108	10	10
GBDT	probability	20261109	4	5
MLP	probability	20261109	7	9
GBDT	threshold	20261109	7	9
MLP	threshold	20261109	10	10
GBDT	probability	20261110	4	6
MLP	probability	20261110	7	9
GBDT	threshold	20261110	8	9
MLP	threshold	20261110	10	10
GBDT	probability	20261111	4	5
MLP	probability	20261111	8	9
GBDT	threshold	20261111	7	7
MLP	threshold	20261111	10	10
GBDT	probability	20261112	4	5
MLP	probability	20261112	7	9
GBDT	threshold	20261112	7	9
MLP	threshold	20261112	10	10

vectors are projected exactly onto the high-degree Walsh subspace with integer transforms. Exact rational evaluation against the released binary64 targets gives certified endpoints, independently of solver status. At cap twelve the identity approximation has zero error exactly. A narrow bracket is not assumed when a solve reaches its time limit.

Because δ_b is nonincreasing, bounds from other budgets tighten each bracket without changing its validity: the lower endpoint at b is the largest lower endpoint over budgets at least b, and its upper endpoint is the smallest upper endpoint over budgets at most b. For each task we report the last budget certified insufficient and the first budget with an approximation residual strictly below 0.025. Budgets between these endpoints are unresolved by the released certificates. Crossing this approximation threshold permits, but does not by itself provide, a finite-sample interval of that half-width.

S7 Empirical Oracle and Evaluation Details

S7.1 Finite-Sample Oracle for Sequential Designs

The fixed-form oracle is a pointwise benchmark. Let $z = (x, y)$, $t(z) = yd(x)$, and let P now denote the held-out empirical law. The oracle may use P to select a design and tables, but the resulting estimator must remain unbiased under every possible law. Otherwise the optimization would trivially return the constant $\Delta(P)$. The admissible score functions $H_\iota(\ell_\iota(x), y)$ therefore satisfy

$$\sum_\iota q_\iota H_\iota(\ell_\iota(x), y) = yd(x) \qquad \text{for every } (x, y). \tag{2}$$

Taking the sample mean gives variance $\{\sum_\iota q_\iota \mathbb{E}_P H_\iota^2 - \Delta(P)^2\}/n$. The optimum of this one-person score problem is also the exact pointwise optimum over all globally unbiased procedures, including those that choose a later person's form using earlier observations. We state the result for arbitrary finite observation maps to make its scope precise.

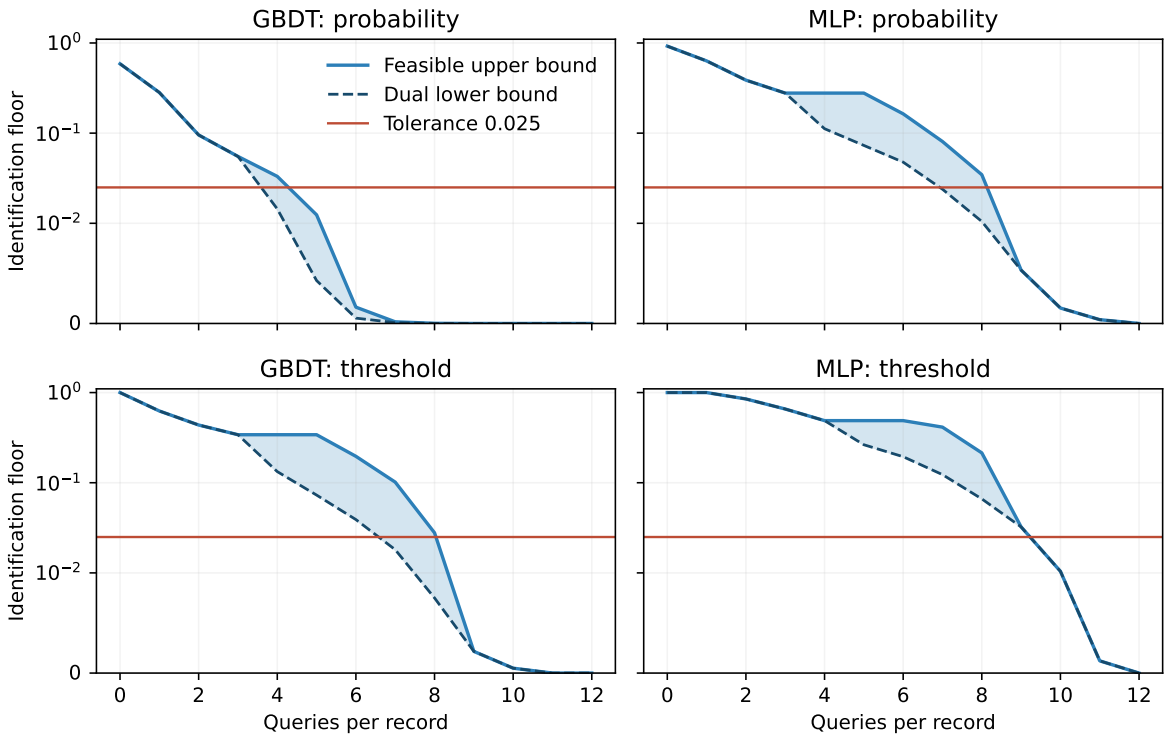


Figure 2: Median certified identification-floor brackets across five fixed WDBC splits. Solid curves are upper bounds, dashed curves lower bounds, and shading shows the remaining uncertainty. The horizontal line is the target 0.025. Upper and lower medians need not come from the same split. The vertical scale is linear below 0.01 and logarithmic above.

Proposition 1 (Pointwise oracle under sequential form choice)**.** *Let Z have an arbitrary law P on a finite space, let $t(Z)$ be known, and let $o_\iota(Z)$ be a finite menu of observation maps whose cells span t. Put $\Delta(P) = \mathbb{E}_P t$ and*

$$W(P) = \inf_{q,H:\,\sum_\iota q_\iota H_\iota(o_\iota(z))=t(z)\ \forall z} \left\{ \sum_\iota q_\iota \mathbb{E}_P H_\iota(o_\iota(Z))^2 - \Delta(P)^2 \right\}. \tag{3}$$

Given n iid records, consider protocols that choose the map for person i before observing any part of Z_i, using all previous observations and independent randomness. Among estimators that are unbiased under every law and have finite second moment under every law,

$$\inf_{\text{such protocols and estimators}} \mathrm{Var}_P(\widehat{\Delta}) = W(P)/n$$

for every P, including laws with zero-mass cells. An iid form mixture and its score mean attain the infimum whenever the program has a minimizer, and otherwise approach it.

Proof. The upper bound follows by using any feasible pair (q, H) independently on the n people. For the converse, first suppose P has full support. For a fixed form law q, let $W_q(P)$ be the value in (6) with q fixed. Write $c = t - \Delta(P)$ and let $\Pi_\iota h = \mathbb{E}_P[h(Z) \mid o_\iota(Z)]$ on the mean-zero space $L_0^2(P)$. A minimum-norm decomposition gives

$$W_q(P) = \sup_{h \in L_0^2(P)} \frac{\langle c, h\rangle_P^2}{\sum_\iota q_\iota \|\Pi_\iota h\|_{2,P}^2}. \tag{4}$$

The value is infinite when the numerator is nonzero and the denominator is zero; $0/0$ contributes zero. To verify the identity, center any feasible score by setting $K_\iota = H_\iota - \Delta(P)$. Its weighted means sum to zero.

Subtracting the individual means $\mathbb{E}_P K_\iota$ from the K_ι preserves $\sum_\iota q_\iota K_\iota = c$ and decreases the objective by $\sum_\iota q_\iota (\mathbb{E}_P K_\iota)^2$. It therefore suffices to minimize $\sum_\iota q_\iota \|K_\iota\|_2^2$ over centered tables with weighted sum c. On the product of these centered transcript spaces, the map $K \mapsto \sum_\iota q_\iota K_\iota$ has adjoint $h \mapsto (\Pi_\iota h)_\iota$ in the weighted inner product. Minimum-norm duality, followed by optimization over the scale of h, gives (7). Forms of mass zero are omitted. If the constraint is infeasible, separation gives a zero-denominator witness, so the extended-value identity remains valid.

Fix any admissible sequential protocol and reveal its independent random seed. For a centered h, use the submodel $P_\theta(z) = P(z)(1 + \theta h(z))$ for sufficiently small $|\theta|$. If person i receives form A_i, the likelihood score contributed by that person's observation at $\theta = 0$ is $\Pi_{A_i} h(Z_i)$. Form-choice probabilities add no likelihood term because the protocol, conditional on its past observations and seed, is fixed rather than a function of the unknown θ. These scores are martingale differences: Z_i is independent of the previous records and $\mathbb{E}_P h = 0$. Thus the score S_h of the entire experiment satisfies

$$\mathbb{E}_P S_h^2 = n \sum_\iota \bar{q}_\iota \|\Pi_\iota h\|_{2,P}^2, \qquad \bar{q}_\iota = \frac{1}{n} \sum_{i=1}^n \Pr_P(A_i = \iota). \tag{5}$$

The probabilities in $\bar{q}$ may depend on P, but $\bar{q}$ is a valid design distribution for the pointwise oracle. Differentiating global unbiasedness along the submodel gives

$$\mathbb{E}_P[(\widehat{\Delta} - \Delta(P)) S_h] = \mathbb{E}_P[ch].$$

Cauchy–Schwarz and (7) then imply

$$\mathrm{Var}_P(\widehat{\Delta}) \geq W_{\bar{q}}(P)/n \geq W(P)/n.$$

This argument allows a nonlinear estimator and arbitrary dependence of later forms on earlier observations.

It remains to treat a law P on the boundary of the simplex. Let $P_\epsilon = (1-\epsilon)P + \epsilon P_0$ for a fixed full-support law P_0. For a fixed protocol, condition on the entire sequence of latent full records and average only over its independent random seed. Since the record space is finite and the estimator has finite second moment under P_0, every resulting conditional second moment is finite. The estimator's first and second moments are therefore finite polynomials in the cell probabilities; in particular its variance is continuous at P. If $F(P) = W(P) + \Delta(P)^2$, the feasible set defining F is population-independent and its objective is linear in the population law for each design. Hence

$$F(P_\epsilon) \geq (1 - \epsilon) F(P), \qquad \liminf_{\epsilon \downarrow 0} W(P_\epsilon) \geq W(P).$$

Apply the already proved lower bound at P_ϵ and pass to the limit. This proves the result at every law. □

For our empirical benchmark, take $Z = (X, Y)$, $t(Z) = Y d(X)$ and $o_T(Z) = (X_T, Y)$. Each fixed form must be chosen before the current person's features or label are seen, as in Section 2. The proposition applies conditionally on the frozen training data. Its oracle chooses a protocol using P, then requires that same protocol and estimator to be unbiased under every alternative law. This permits a distribution-specific benchmark without allowing the trivial constant estimator $\Delta(P)$. The lower bound covers predetermined form counts and adaptation across respondents. It does not cover estimators that are biased at other populations, such as the fitted imputation comparator.

S7.2 Computing and Evaluating the Oracle

All three-feature-and-label marginals used in the numerical solvers were positive, although the proposition also permits empty full-profile cells.

Taking the sample mean gives variance $\{\sum_\iota q_\iota \mathbb{E}_P H_\iota^2 - \Delta(P)^2\}/n$. The optimum of this one-person score problem is also the exact pointwise optimum over all globally unbiased procedures, including those that choose a later person's form using earlier observations. We state the result for arbitrary finite observation maps to make its scope precise.

Proposition 2 (Pointwise oracle under sequential form choice)**.** *Let Z have an arbitrary law P on a finite space, let $t(Z)$ be known, and let $o_\iota(Z)$ be a finite menu of observation maps whose cells span t. Put $\Delta(P) = \mathbb{E}_P t$ and*

$$W(P) = \inf_{q,H:\,\sum_\iota q_\iota H_\iota(o_\iota(z))=t(z)\ \forall z} \left\{ \sum_\iota q_\iota \mathbb{E}_P H_\iota(o_\iota(Z))^2 - \Delta(P)^2 \right\}. \tag{6}$$

Given n iid records, consider protocols that choose the map for person i before observing any part of Z_i, using all previous observations and independent randomness. Among estimators that are unbiased under every law and have finite second moment under every law,

$$\inf_{\text{such protocols and estimators}} \mathrm{Var}_P(\widehat{\Delta}) = W(P)/n$$

for every P, including laws with zero-mass cells. An iid form mixture and its score mean attain the infimum whenever the program has a minimizer, and otherwise approach it.

Proof. The upper bound follows by using any feasible pair (q, H) independently on the n people. For the converse, first suppose P has full support. For a fixed form law q, let $W_q(P)$ be the value in (6) with q fixed. Write $c = t - \Delta(P)$ and let $\Pi_\iota h = \mathbb{E}_P[h(Z) \mid o_\iota(Z)]$ on the mean-zero space $L_0^2(P)$. A minimum-norm decomposition gives

$$W_q(P) = \sup_{h \in L_0^2(P)} \frac{\langle c, h\rangle_P^2}{\sum_\iota q_\iota \|\Pi_\iota h\|_{2,P}^2}. \tag{7}$$

The value is infinite when the numerator is nonzero and the denominator is zero; $0/0$ contributes zero. To verify the identity, center any feasible score by setting $K_\iota = H_\iota - \Delta(P)$. Its weighted means sum to zero. Subtracting the individual means $\mathbb{E}_P K_\iota$ from the K_ι preserves $\sum_\iota q_\iota K_\iota = c$ and decreases the objective by $\sum_\iota q_\iota (\mathbb{E}_P K_\iota)^2$. It therefore suffices to minimize $\sum_\iota q_\iota \|K_\iota\|_2^2$ over centered tables with weighted sum c. On the product of these centered transcript spaces, the map $K \mapsto \sum_\iota q_\iota K_\iota$ has adjoint $h \mapsto (\Pi_\iota h)_\iota$ in the weighted inner product. Minimum-norm duality, followed by optimization over the scale of h, gives (7). Forms of mass zero are omitted. If the constraint is infeasible, separation gives a zero-denominator witness, so the extended-value identity remains valid.

Fix any admissible sequential protocol and reveal its independent random seed. For a centered h, use the submodel $P_\theta(z) = P(z)(1 + \theta h(z))$ for sufficiently small $|\theta|$. If person i receives form A_i, the likelihood score contributed by that person's observation at $\theta = 0$ is $\Pi_{A_i} h(Z_i)$. Form-choice probabilities add no likelihood term because the protocol, conditional on its past observations and seed, is fixed rather than a function of the unknown θ. These scores are martingale differences: Z_i is independent of the previous records and $\mathbb{E}_P h = 0$. Thus the score S_h of the entire experiment satisfies

$$\mathbb{E}_P S_h^2 = n \sum_\iota \bar{q}_\iota \|\Pi_\iota h\|_{2,P}^2, \qquad \bar{q}_\iota = \frac{1}{n} \sum_{i=1}^n \Pr_P(A_i = \iota). \tag{8}$$

The probabilities in $\bar{q}$ may depend on P, but $\bar{q}$ is a valid design distribution for the pointwise oracle. Differentiating global unbiasedness along the submodel gives

$$\mathbb{E}_P[(\widehat{\Delta} - \Delta(P)) S_h] = \mathbb{E}_P[ch].$$

Cauchy–Schwarz and (7) then imply

$$\mathrm{Var}_P(\widehat{\Delta}) \geq W_{\bar{q}}(P)/n \geq W(P)/n.$$

This argument allows a nonlinear estimator and arbitrary dependence of later forms on earlier observations.

It remains to treat a law P on the boundary of the simplex. Let $P_\epsilon = (1-\epsilon)P + \epsilon P_0$ for a fixed full-support law P_0. For a fixed protocol, condition on the entire sequence of latent full records and average only over its independent random seed. Since the record space is finite and the estimator has finite second moment under P_0, every resulting conditional second moment is finite. The estimator's first and second moments are therefore finite polynomials in the cell probabilities; in particular its variance is continuous at P. If

$F(P) = W(P) + \Delta(P)^2$, the feasible set defining F is population-independent and its objective is linear in the population law for each design. Hence

$$F(P_\epsilon) \geq (1-\epsilon)F(P), \qquad \liminf_{\epsilon \downarrow 0} W(P_\epsilon) \geq W(P).$$

Apply the already proved lower bound at P_ϵ and pass to the limit. This proves the result at every law. □

For our empirical benchmark, take $Z = (X, Y)$, $t(Z) = Yd(X)$ and $o_T(Z) = (X_T, Y)$. Each fixed form must be chosen before the current person's features or label are seen, as in Section 2. The proposition applies conditionally on the frozen training data. Its oracle chooses a protocol using P, then requires that same protocol and estimator to be unbiased under every alternative law. This permits a distribution-specific benchmark without allowing the trivial constant estimator $\Delta(P)$. The lower bound covers predetermined form counts and adaptation across respondents. It does not cover estimators that are biased at other populations, such as the fitted imputation comparator.

S7.3 Computing and Evaluating the Oracle

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S7.4 One-Sided Deterministic Updates Can Have an Exponential Gap

Large global gaps do not require complementary classifiers. Let A have $\kappa \geq 1$ address bits, let $D = 2^\kappa$, and write $Z_j \in \{0, 1\}$ for its data bits. The old rule always predicts -1, and the new rule predicts $+1$ exactly when $Z_A = 1$. Their contrast is $d = Z_A \in \{0, 1\}$.

Proposition 3. *For this contrast, $m = \kappa + D$, $b = \kappa + 1$, and*

$$V_{\text{branch}}(d, b) = 1, \qquad V_{\text{fixed}}(d, b) = \frac{D^2}{4(D-1)}.$$

The ratio is asymptotic to 2^{b-3}. Multiplying the contrast by any $\epsilon \in (0, 1]$ leaves the ratio unchanged.

Proof. Put $\pi = D/[2(D-1)]$. Choose a data index J uniformly, query all address bits and Z_J, and return the feature score

$$H = \pi + D\mathbf{1}\{J = A\}(Z_J - \pi).$$

Its conditional mean is d. If $J = A$, its value is $D(d - 1/2)$; otherwise it is π. Hence its conditional second moment is $D/4 + (D-1)\pi^2/D = D^2/[4(D-1)]$ for either value of d.

For the converse, let μ have a uniform address and independent Bernoulli(π) data bits, and take $h = \pi + D(d - \pi)$. A fixed form with s address bits and r data bits has

$$\|\mathbb{E}_\mu[h \mid X_T]\|_\mu^2 = \pi^2 + \pi(1-\pi)r2^s \leq \pi^2 + D\pi(1-\pi) =: F,$$

because $r + s \leq \kappa + 1$ implies $r2^s \leq D$. Also $\langle d, h \rangle_\mu = D\pi - (D-1)\pi^2 = F$. The variance dual therefore gives $V_{\text{fixed}} \geq F = D^2/[4(D-1)]$. This calculation includes $D = 2$, where $\pi = 1$ and the lower bound is simply one. Branching evaluates d exactly, and Jensen gives its matching coefficient one. Both coefficients scale by ϵ^2. □

This is a constructed model family, not a claim about how frequently trained classifiers have such gaps. It separates the structural question from model-family labels: a GBDT or neural-network name alone determines neither identifiability nor the global efficiency ratio.

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